

Q- Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.

To load a pre-configured Cisco Packet Tracer network containing three routers configured with RIP and use the following commands to observe the routing information:

- show ip route
- show ip protocols

The purpose is to identify RIP-learned routes and understand their metrics.

1. Open **Cisco Packet Tracer**.
2. Open the provided pre-configured file: **first.pkt**
3. Verify that three routers are present and connected.
4. Wait for the router interfaces and RIP routing information to converge.

Step 2: Access Router R1

Click Router R1 and open the **CLI**.

Enter:

```
enable
show ip route
```

The routing table displays directly connected routes and routes learned through RIP.

RIP routes are identified by the letter:

R

For example:

```
R 192.168.2.0/24 [120/1] via 10.0.12.2
R 192.168.3.0/24 [120/2] via 10.0.12.2
```

Here:

- R = Route learned through RIP
- 120 = Administrative Distance of RIP
- 1 or 2 = RIP metric/hop count
- via = Next-hop router

Step 3: Check RIP Configuration on R1

Enter:

```
show ip protocols
```

The command displays information about the active routing protocol.

Typical output includes:

```
Routing Protocol is "rip"  
Sending updates every 30 seconds  
Invalid after 180 seconds  
Hold down 180  
Flush after 240
```

It also displays the networks participating in RIP and information about neighboring routers.

Step 4: Repeat on Router R2

Enter:

```
enable  
show ip route
```

R2 should show routes learned from R1 and R3.

Example:

```
R 192.168.1.0/24 [120/1] via 10.0.12.1  
R 192.168.3.0/24 [120/1] via 10.0.23.3
```

Then enter:

```
show ip protocols
```

This confirms that RIP is running on R2.

Step 5: Repeat on Router R3

Enter:

```
enable  
show ip route
```

Example:

```
R 192.168.1.0/24 [120/2] via 10.0.23.2  
R 192.168.2.0/24 [120/1] via 10.0.23.2
```

Then enter:

show ip protocols

This confirms that RIP is configured and exchanging routing updates.

6. Observation Table

Router	RIP-Learned Network	Next Hop	Metric
R1	LAN behind R2	R2	1 hop
R1	LAN behind R3	R2	2 hops
R2	LAN behind R1	R1	1 hop
R2	LAN behind R3	R3	1 hop
R3	LAN behind R2	R2	1 hop
R3	LAN behind R1	R2	2 hops

Important Output Interpretation

For a routing-table entry such as:

R 192.168.3.0/24 [120/2] via 10.0.12.2

the meaning is:

Part	Meaning
R	Route was learned using RIP
192.168.3.0/24	Destination network
120	RIP administrative distance
2	RIP metric
via 10.0.12.2	Next-hop router

The **RIP metric** is measured in hop counts.

For example:

- Metric **1** = destination is one router hop away.

- Metric **2** = destination is two router hops away.
- Metric **3** = destination is three router hops away.

RIP selects the route with the **lowest hop count**.

In this experiment, RIP routing was observed on a three-router Cisco Packet Tracer network. The `show ip route` command helped identify RIP-learned routes and their metrics, while `show ip protocols` provided details about the RIP configuration.

It was observed that RIP uses **hop count as its metric**, has a default **administrative distance of 120**, and considers a maximum of **15 hops** as reachable. A metric of 16 indicates an unreachable destination.

Therefore, the experiment successfully demonstrated how routers learn remote networks dynamically using **RIP (Routing Information Protocol)**.